

Gaze-Controlled Dual-Arm Teleoperation Based on Visual Separation Scheme: Towards Operational Space-Sensitive Elderly Assistive Manipulation

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Abstract—Global population ageing is increasing the demand for teleoperated assistive robots in domestic environments. Elderly-care scenarios often involve *operational space sensitivity* (OSS), where confined layouts, cluttered objects, and residual human mobility tighten coordination and safety constraints. In such settings, operators must alternate between issuing manipulation commands and monitoring dynamic surroundings, leading to visual functional conflict. To address this, we propose a *Visual Separation Scheme* (VSS) that structures visual allocation for command execution and situational awareness. Based on VSS, a gaze-controlled dual-arm shared autonomy framework is developed: the primary arm follows proportional gaze-to-velocity mapping, while the assisting arm executes cooperative behaviour through a state-conditioned policy. The framework is evaluated in OSS-constrained assistive tasks using completion time, placement error, trajectory smoothness, and collision count.

Index Terms—Gaze teleoperation, Dual-arm manipulation, Shared autonomy, Assistive robotics, Operational space sensitivity

I. INTRODUCTION

Gaze-based teleoperation leverages visual attention as a direct and hands-free input modality. Dual-arm robotic systems are commonly coordinated through model-based planning and feedback regulation [1], [2], which have been shown to reduce operator cognitive load in complex tasks [3]. With ageing societies, remote dual-arm robots are increasingly explored for domestic rehabilitation and assistive manipulation. Unlike industrial environments, elderly users often retain partial mobility, requiring robots to operate accurately while maintaining safe and predictable interaction in shared living spaces.

A central challenge in these environments is *operational space sensitivity* (OSS). Redundant dual-arm robots operating in open environments must address spatial constraints and real-time collision avoidance [4]. OSS reflects layered constraints arising from limited physical scale, partially unpredictable human motion, and strict interactive safety margins. In this work, OSS is characterised through three dimensions: geometric sensitivity (dense layouts and restricted motion corridors), uncertainty sensitivity (variable user motion), and interaction sensitivity (the need to preserve a non-intrusive safety buffer). Under these conditions, single-arm manipulation may struggle

to ensure both stability and avoidance, whereas coordinated dual-arm control improves support and safety margins. Fig. 1 illustrates these sensitivity dimensions.

Meanwhile, operators must generate precise control commands while continuously monitoring environmental dynamics, intensifying visual resource allocation conflict. To mitigate this issue, we introduce a structured Visual Separation Scheme (VSS) that explicitly allocates visual resources between command generation and situational monitoring. Based on this mechanism, we develop a gaze-controlled dual-arm shared autonomy framework tailored for OSS elderly assistive manipulation.

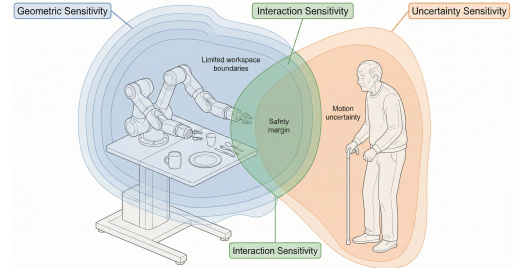


Fig. 1. OSS for elderly assistive environments.

II. VSS-BASED DUAL-ARM FRAMEWORK

A. Visual Separation Schedule

We define a schedule that allocates visual resources between command generation and monitoring. Fig. 2 presents the visual separation mechanism, where foveal vision is prioritised for command generation while peripheral vision supports situational awareness. Let $w_f(t) \in [0, 1]$ denote the proportion allocated to foveal command generation and $w_p(t) = 1 - w_f(t)$ to peripheral situational monitoring.

The primary arm velocity is generated through proportional mapping between gaze offset and end-effector velocity:

$$\mathbf{v}_R(t) = k(\mathbf{g}(t) - \mathbf{c}) \quad (1)$$

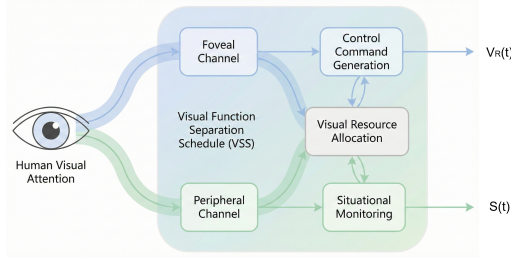


Fig. 2. VSS mechanism.

where $\mathbf{g}(t)$ denotes the gaze point, $\mathbf{c}$ represents the neutral reference position, and k is a scaling gain. The assisting arm velocity is generated by a state-conditioned policy:

$$\mathbf{v}_L(t) = f_{\theta}(\mathbf{s}(t), \mathbf{v}_R(t)) \quad (2)$$

where $\mathbf{s}(t)$ includes primary-arm velocity, relative pose, object state, and distances to surrounding entities. The cooperative policy is trained using a weighted multi-objective loss:

$$\mathcal{L} = \alpha \mathcal{L}_{coord} + \beta \mathcal{L}_{avoid} + \gamma \mathcal{L}_{smooth}, \quad (3)$$

where $\mathcal{L}_{coord}$ measures bimanual coordination error, $\mathcal{L}_{avoid}$ penalises violations of safety distance/obstacle proximity, and $\mathcal{L}_{smooth}$ enforces motion smoothness. The weights are selected on a validation task: the three sub-objectives are normalized, and a grid search is performed to choose (α, β, γ) that best balances coordination stability, safety distance maintenance, and trajectory quality, subject to $\alpha + \beta + \gamma = 1$.

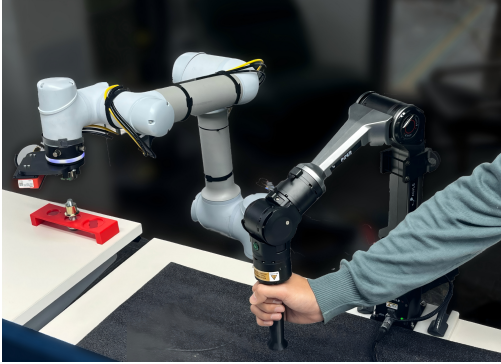


Fig. 3. Experimental setup of the gaze-controlled dual-arm teleoperation.

III. EXPERIMENTAL DESIGN

A. Task Definition

We evaluate the framework on an operational space-sensitive assistive grasp-and-place task inspired by domestic elderly-care scenarios. The experimental configuration is shown in Fig. 3, including the gaze input module, dual-arm robotic platform, and environmental obstacle arrangement. The task involves dual-point support for large or heavy objects, continuous avoidance of surrounding obstacles, and precise placement near a target region while preserving safe interaction space for a nearby user. Due to OSS and potential human motion, coordinated dual-arm execution is required.

B. Baselines

We compare against (i) single-arm gaze control, (ii) dual-arm control without a cooperative policy, and (iii) the proposed VSS-based dual-arm shared autonomy framework.

C. Evaluation Metrics

Performance is evaluated in terms of efficiency, accuracy, motion quality, and safety. Efficiency is measured by completion time T . Accuracy is quantified as placement error $E = \|\mathbf{x}_{final} - \mathbf{x}_{target}\|$. Motion quality is evaluated using the acceleration-energy functional $J = \int_0^T \|\ddot{\mathbf{x}}(t)\|^2 dt$. In addition, the total number of collision events during execution is recorded as a safety indicator.

IV. KEY RESULTS / EXPECTED OUTCOMES

We expect the proposed VSS-based dual-arm framework to reduce visual functional conflict by structuring visual allocation between command generation and monitoring. Compared to baselines, the approach is expected to improve stability for large/heavy-object handling while maintaining safe interaction margins in operational space-sensitive settings, leading to fewer collisions and smoother trajectories without sacrificing placement accuracy.

V. CONCLUSION

This paper presented a VSS-guided dual-arm gaze teleoperation framework for OSS elderly assistive manipulation. By explicitly addressing operational space sensitivity and structuring visual allocation between command generation and situational monitoring, the proposed approach balances intuitive operator control with cooperative assisting-arm behavior learned from demonstrations. The framework is evaluated using task-appropriate quantitative metrics and provides a structured approach for scalable dual-arm assistive teleoperation.

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